Trainings

General Information

The purpose of the variable geometry turbocharger:

- When the EGR flow is commanded, the variable geometry turbocharger will close the nozzle in the turbine housing to create more back pressure in the exhaust manifold, forcing the EGR flow back into the engine.
- Closing the variable geometry turbocharger will also increase exhaust gas temperature under certain normal engine operating conditions and during aftertreatment regeneration events.
- The turbocharger also functions to improve engine performance by building boost more quickly during acceleration.

Use the following procedure for further information regarding variable geometry turbocharger and aftertreatment system interactions. Refer to Procedure 011-999 in Section F. (/qs3/pubsys2/xml/en/procedures/35/35-011-999-tr.html)

This procedure provides service information for the electronically controlled variable geometry turbocharger actuators.

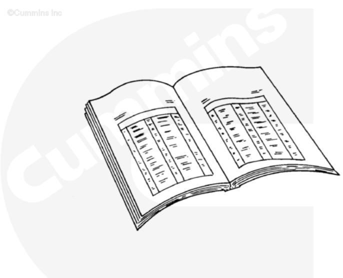
Note : To avoid improper installation of the turbocharger actuator, follow each step in its entirety and in the proper sequence.

The starting motor wiring can electrically interfere with the VGT pigtail and wiring harness. The starting motor wiring **must** be at least 152 mm [6 in] away from the actuator and harness.

Preparatory Steps

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



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⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

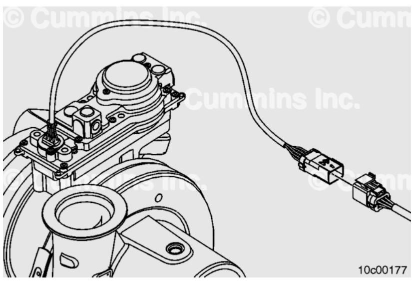
⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Clean the area around the turbocharger actuator and dry with compressed air.
- Drain the engine coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Disconnect the turbocharger actuator coolant supply and return lines from the turbocharger actuator. Refer to Procedure 010-041 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-041-tr.html)
- Remove the turbocharger actuator wiring harness zip ties.

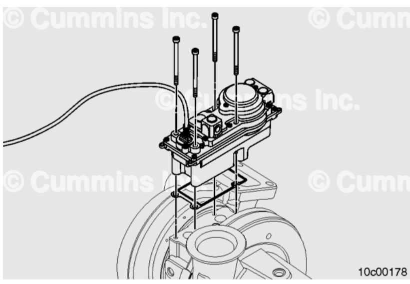
Remove

Disconnect the wiring harness from the turbocharger actuator by sliding the locking tang to the open position, and then pushing down on the release lever and pulling the connection apart.



⚠ CAUTION ⚠

Make sure the coolant has been drained before the actuator is removed. Failure to do so can result in damage to the actuator.



Remove the four turbocharger actuator mounting capscrews with a 5-mm Allen™ wrench and discard the capscrews.

Remove the actuator.

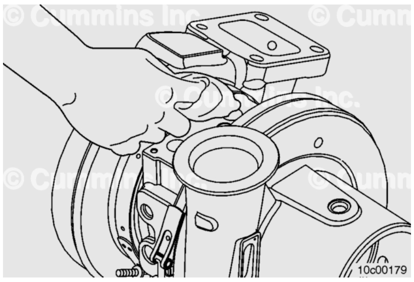
Remove and discard the turbocharger actuator sealing gasket.

Clean and Inspect for Reuse

Use a clean cloth to wipe around the surface where the turbocharger actuator seals to the turbocharger.

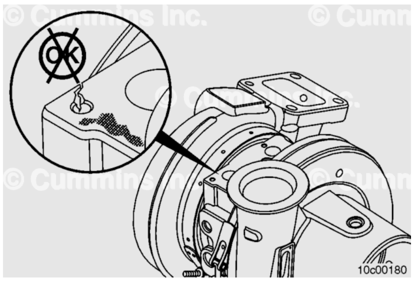
Note : Take care not to drop any dirt or debris into the turbocharger.

Inspect the sealing surfaces and the sealing gasket for signs of leaks, cracks, or other damage.



If the sealing surface of the turbocharger is damaged and causing a leak, the turbocharger **must** be replaced. Discontinue work with this procedure. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)

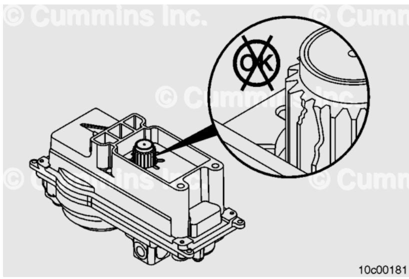
If the sealing surface of the actuator is causing a leak, replace **only** the actuator.



Inspect the pinion gear on the actuator for seizure, excessive wear, or damaged teeth.

Confirm the actuator is disconnected from the engine harness. Rotate the output pinion gear by hand. Do **not** use any tools to turn the gear.

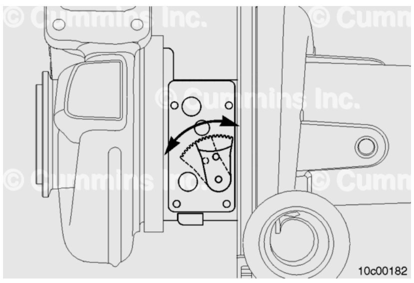
Replace the actuator if the pinion gear is seized, or if the teeth are worn or damaged.



Inspect the sector gear on the turbocharger for excessive wear, damaged teeth, or a broken shaft.

Grasp the sector gear by hand and move it through its operational range.

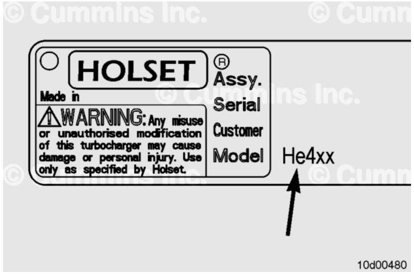
Within the turbocharger, there are solid end stops, which limit sector gear travel in both directions.



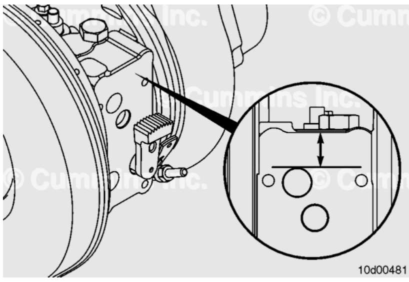
The sector gear **must** move smoothly by hand through the entire range of motion. It takes considerable effort to begin moving the sector gear. However, once the sector gear begins to move, minimal force is required to continue moving the sector gear through its operating range.

The sector gear travel gages are designed for specific turbocharger models. To determine the correct gage, verify the turbocharger model number from the turbocharger data plate. Select the gage that matches the first three letters of the turbocharger model number.

Example: HE5xx, HE4xx, and so on.

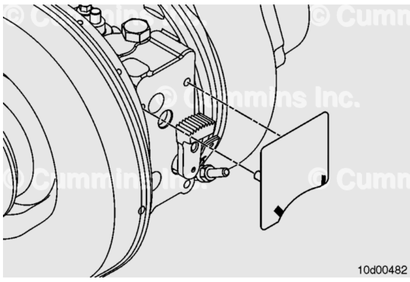


An alternate method of determining the turbocharger model number, if the turbocharger data tag is missing or covered with paint, is to measure the distance between the top of the cup plug hole to the bearing housing edge. If the distance is greater than 19 mm [0.748 in], the turbocharger model is HE4xx. If the distance is less than 19 mm [0.748 in], the model is HE5xx.

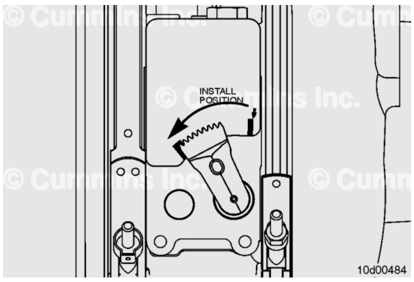


To install the sector gear travel gage, carefully bend the gage and slide the thin section under the sector gear. If necessary, pull the sector gear out by hand to allow more clearance for the gage.

Verify that the three alignment bosses are fully engaged in the bearing housing.



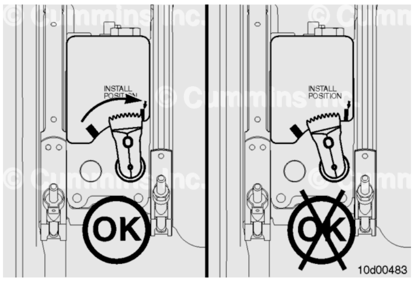
Rotate the sector gear **counterclockwise**, toward the turbocharger turbine housing, until it stops. The edge of the sector gear **must** be in the green acceptance zone of the gage.



Rotate the sector gear **clockwise**, toward the turbocharger compressor housing, until it stops. The edge of the sector gear **must** be in the green acceptance zone of the gage.

If the sector gear does **not** go through the full range of motion, replace the turbocharger.

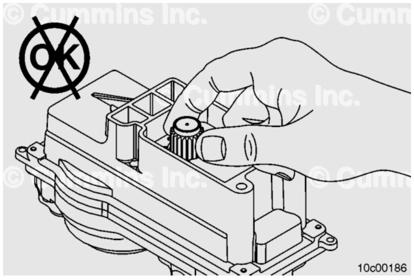
Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)



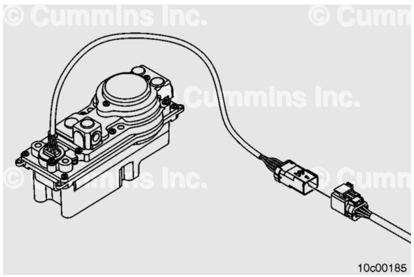
Install

⚠ **CAUTION** ⚠

During the turbocharger actuator install procedure, the pinion gear on the actuator will move. Keep hands and tools away from the pinion gear during the procedure.



Following these instructions in order is very important.



- Continue through the entire turbocharger actuator installation procedure before attempting to troubleshoot any other fault codes.
- Verify that the turbocharger actuator is removed from the turbocharger bearing housing.
- Verify that the turbocharger actuator electrical connector is disconnected from the engine wiring harness.
- Turn the keyswitch ON. Connect INSITE™ electronic service tool and wait 60 seconds.

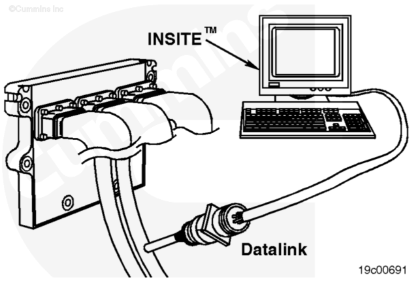
  Note : It is important that the keyswitch remain in the ON position throughout both the Install and Calibrate procedures unless otherwise directed.

- Connect the turbocharger actuator electrical connector to the engine wiring harness.

If Fault Code 2634 becomes active, disconnect the turbocharger actuator connector from the engine wiring harness with the keyswitch ON. Connect the turbocharger actuator electrical connector. Fault Code 2634 will go inactive.

It is normal and expected to have Fault Code 2449 active when a new turbocharger actuator is connected to the engine, because it is **not** calibrated to the turbocharger.

Continue through the engine turbocharger actuator installation procedure before attempting to troubleshoot any other fault codes.



Use INSITE™ electronic service tool and go to the ECM Diagnostic Tests screen.

From the list, select VGT Electronic Actuator Installation and Calibration and click on the “next” button.

Note : The VGT Electronic Actuator Installation and Calibration is not a diagnostic test. It is the procedure to properly install and calibrate the turbocharger actuator. Running this procedure improperly can result in additional fault codes and/or damage to the turbocharger or actuator.

The INSITE™ electronic service tool “INSTALL ACTUATOR” command **must only** be run with the actuator **not** mounted to the turbocharger.

Locate the column labeled “Value” and left click on the down arrow. Select “INSTALL ACTUATOR” and select “START”.

This will set the actuator pinion gear to a known position to prepare it for installation to the turbocharger. This step should take less than 30 seconds to complete with INSITE™ electronic service tool.

INSITE™ electronic service tool will indicate when this step is complete.

Fault Code 2449 will be active at this point in the procedure. Continue through the turbocharger actuator procedure before troubleshooting fault codes.

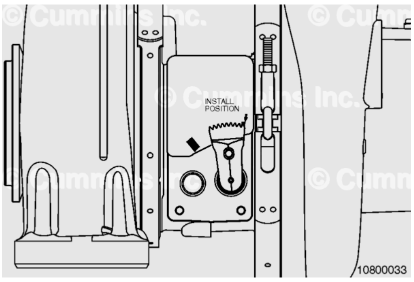
If at any point, INSITE™ electronic service tool status message indicates the procedure was stopped or failed, leave the keyswitch ON and cycle the power to the actuator by unplugging it from the harness, and then reconnecting it.

If cycling the power to the actuator does **not** work, unplug the actuator, turn the keyswitch OFF for 30 seconds, and then restart INSITE™ electronic service tool. Then start over, beginning with the “INSTALL ACTUATOR” step.

Grasp the sector gear by hand and rotate the sector gear toward the turbocharger compressor housing. Make sure the edge of the sector gear is rotated all the way towards the “Install Position” arrow.

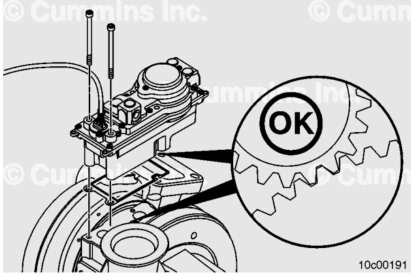
Coat the teeth on the sector gear with the grease packet supplied in the installation kit.

Note : It is critical for smooth reliable operation of the actuator to use the full amount of the Holset® supplied grease.



⚠ CAUTION ⚠

Do not attempt to force the actuator onto the bearing housing by using the capscrews. Misalignment can cause damage to the actuator or turbocharger.



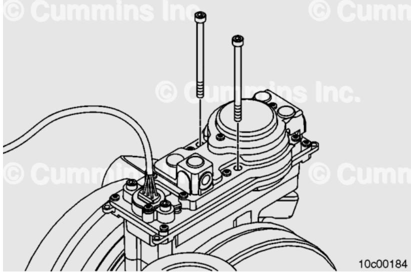
Align the actuator and gasket assembly with the turbocharger bearing housing face with the two lower capscrews. Be sure to use the new capscrews available in the turbocharger actuator mounting kit.

Align the actuator to the bearing housing and slide the actuator into place. The actuator pinion gear and the turbocharger sector gear **must** engage smoothly. If **not**, verify that the sector gear through-hole and the bearing housing blind-hole are aligned.

It could be necessary to twist the actuator housing to align it with the mounting holes.

With the actuator aligned on the bearing housing, install the remaining two capscrews.

Tighten the four capscrews in a crisscross pattern.



Torque Value:

1. 3 n•m [27 in-lb]
2. 11 n•m [97 in-lb]

Calibrate

The turbocharger actuator **must** be calibrated to the turbocharger. This step **must** be performed to make sure proper turbocharger operation is achieved.

Calibration instructions:

Within the INSITE™ electronic service tool screen labeled “VGT Electric Actuator Install and Calibrate”, locate the column labeled “Value” and left click on the down arrow. Select “Calibrate Actuator”.

INSITE™ electronic service tool “CALIBRATE ACTUATOR” command **must only** be run with the actuator mounted to the turbocharger.

Follow the instruction on the screen in order to calibrate the turbocharger actuator to the turbocharger. INSITE™ electronic service tool will indicate when this step is complete.

If INSITE™ electronic service tool status message indicates the procedure was stopped or failed, turn the keyswitch OFF for 30 seconds and key back ON. Then start over, beginning with the “INSTALL ACTUATOR”

It is normal to have an active Fault Code 2387 at this point.

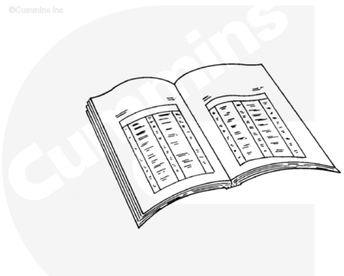
Turn the keyswitch to the OFF position for 30 seconds. Then turn the keyswitch to the ON position and refresh the fault code screen. All turbocharger actuator fault codes should be inactive. Use INSITE™ electronic service tool to clear all fault codes.

Note : Make sure all fault codes are inactive and cleared before adding coolant to the engine.

Finishing Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



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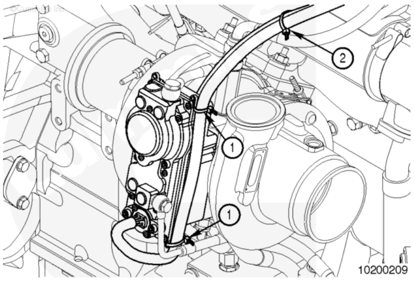
- Install the turbocharger actuator coolant supply and return lines. Refer to Procedure 010-041 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-041-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Operate the engine to make sure the turbocharger actuator operates properly and all turbocharger actuator

fault codes are inactive.
- Check the engine for leaks.

The starting motor wiring can electrically interfere with the VGT pigtail and wiring harness. The starting motor wiring **must** be at least 152 mm [6 in] away from the actuator and harness.

Secure the turbocharger actuator wiring away from hot and abrasive parts.

- Zip tie the turbocharger actuator harness to the two ears on the actuator housing (1).
- Zip tie the turbocharger actuator wire harness to the turbocharger speed sensor wire (2).



Last Modified: 12-Jul-2017